

Causal Diagrams & Model Planning

June 2025

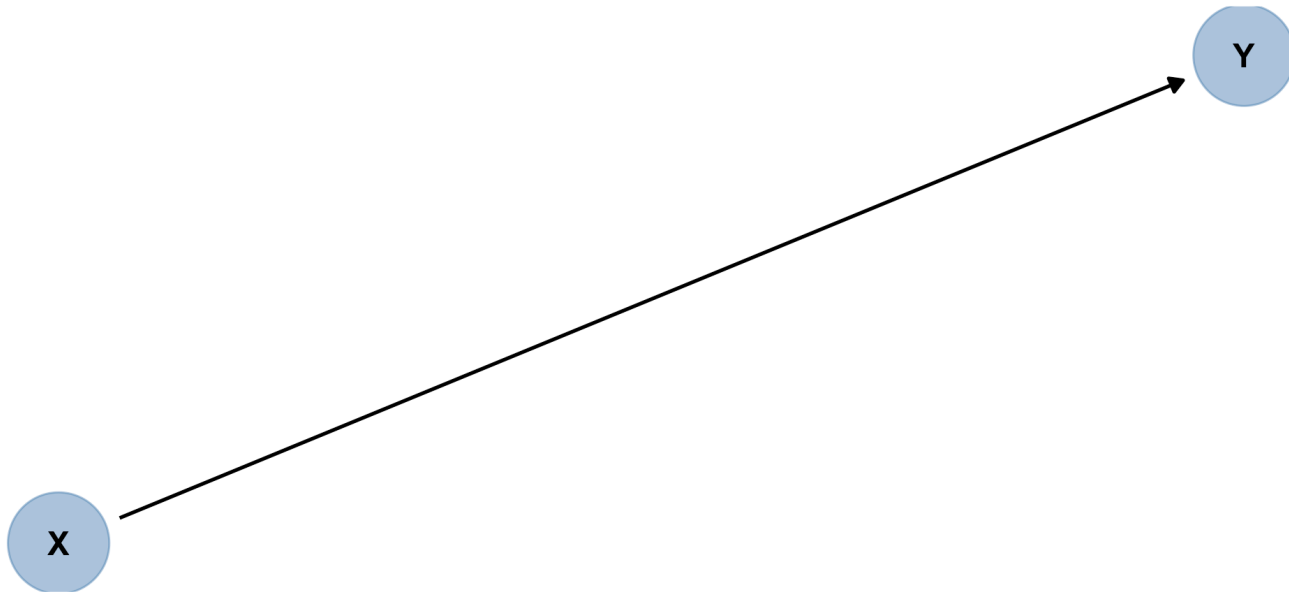
Old System:
**"Include every
predictor you think is
relevant and
important"**

(Data) Size Matters!

- $p < \frac{n}{15}$: Roughly 10-20 "events" per parameter estimated ([Harrell 2015](#))
- Assumes even dist. over categories
- Consider: domain expertise, data availability, missing-ness, dynamic range...& *expected causal relationships*

Causal Diagrams

There's more to planning than just $p < n/15$!



"Causal"?

Can we actually make *causal conclusions*?

- If detect association, can conclude ____ *causes* ____ IFF...
- Otherwise:
- So why use "causal" diagrams to plan models?
 - Transparency about assumptions
 - Clear, consistent way to choose predictors

H.E.A.L.T.H. Camp Example



Health Education And Leadership Training for a Hopeful future (H.E.A.L.T.H) Camp "is a free weeklong summer camp for young people from mostly medically underserved communities in Grand Rapids to learn more about their own health and health careers at Calvin U."

H.E.A.L.T.H. Camp Example

Response: Test Score

- Age
- Sex
- Grade in school
- Neighborhood
- Race & Ethnicity
- Income
- Interest in health career
- Timepoint (before/after camp)

Classify to Plan

- *First* choose a key predictor whose effect you'll quantify
- Determine other candidate predictors' relative roles
- Include/exclude in model depending on this classification

Confounder

a.k.a. "common cause"

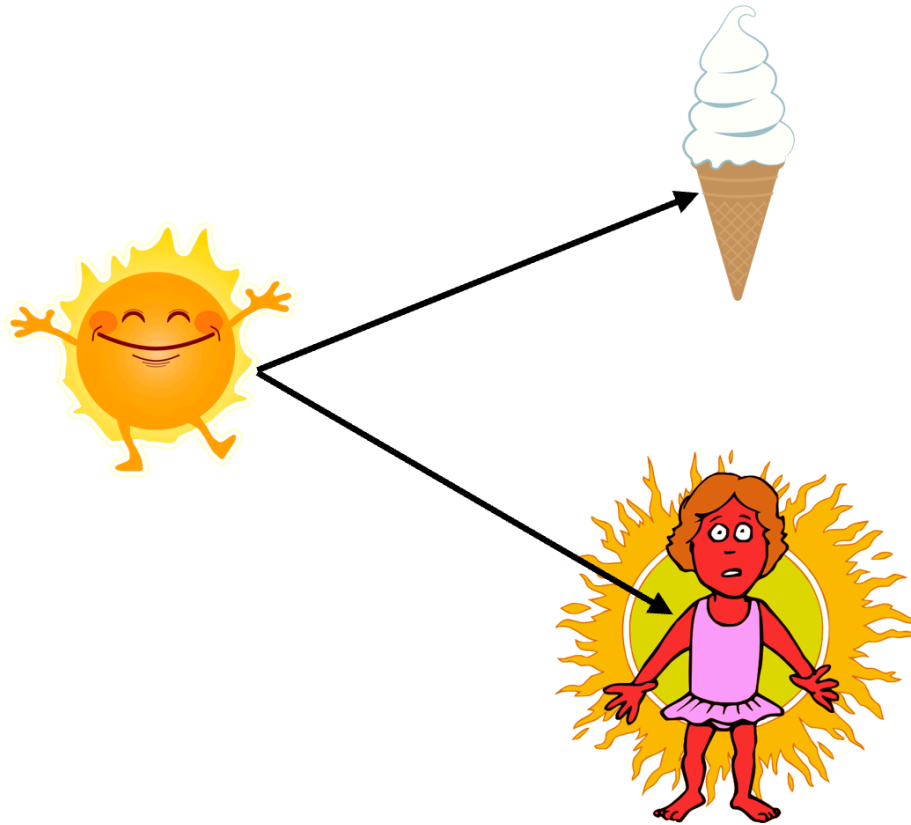
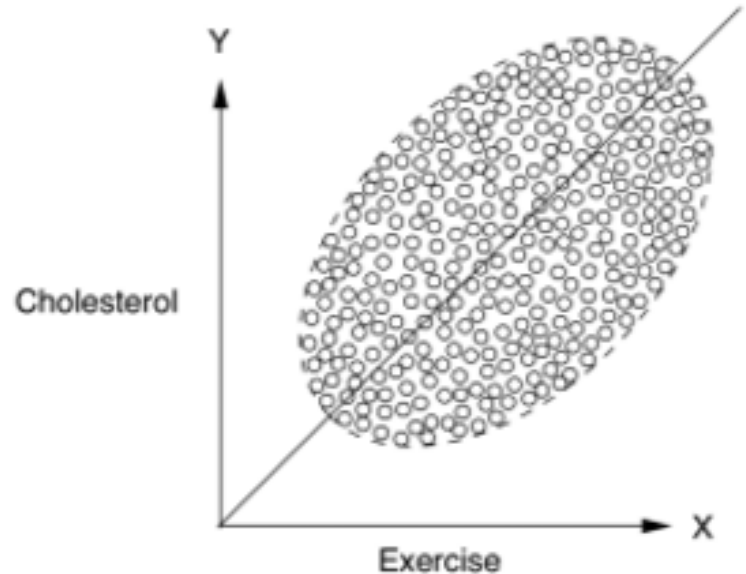
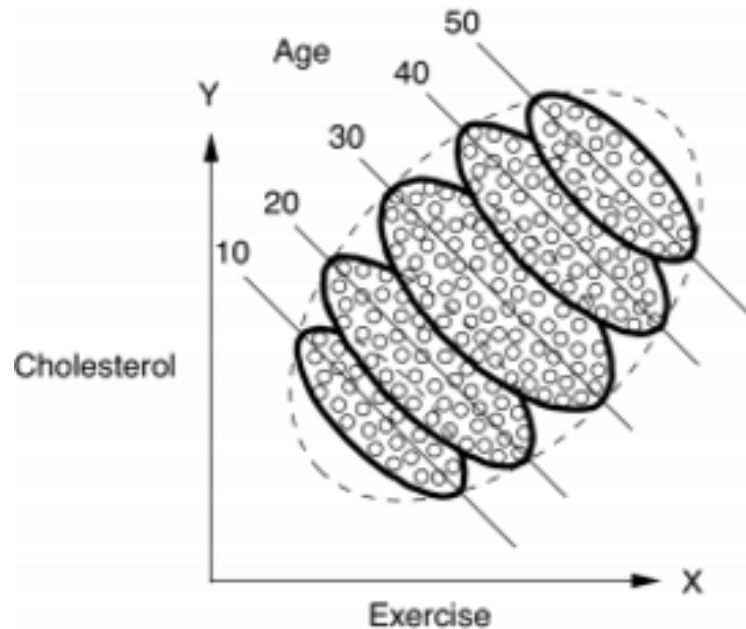


image: [Fieberg, Statistics for Ecologists Figure 7.3](#), used under [CC BY 4.0 license](#)

Confounder



But...What's *wrong* with this picture?

Image: <https://www.causa.tech>

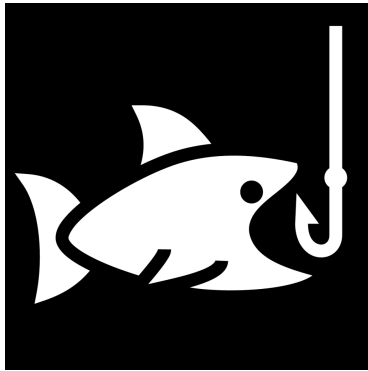
Precision Covariate

Mediator

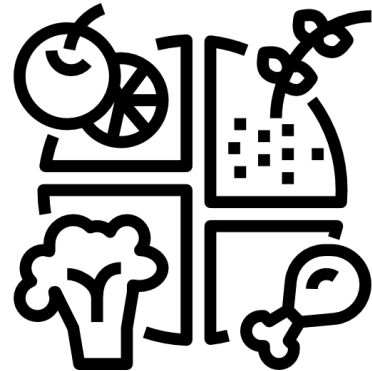
Collider



U. MN St Paul Campus



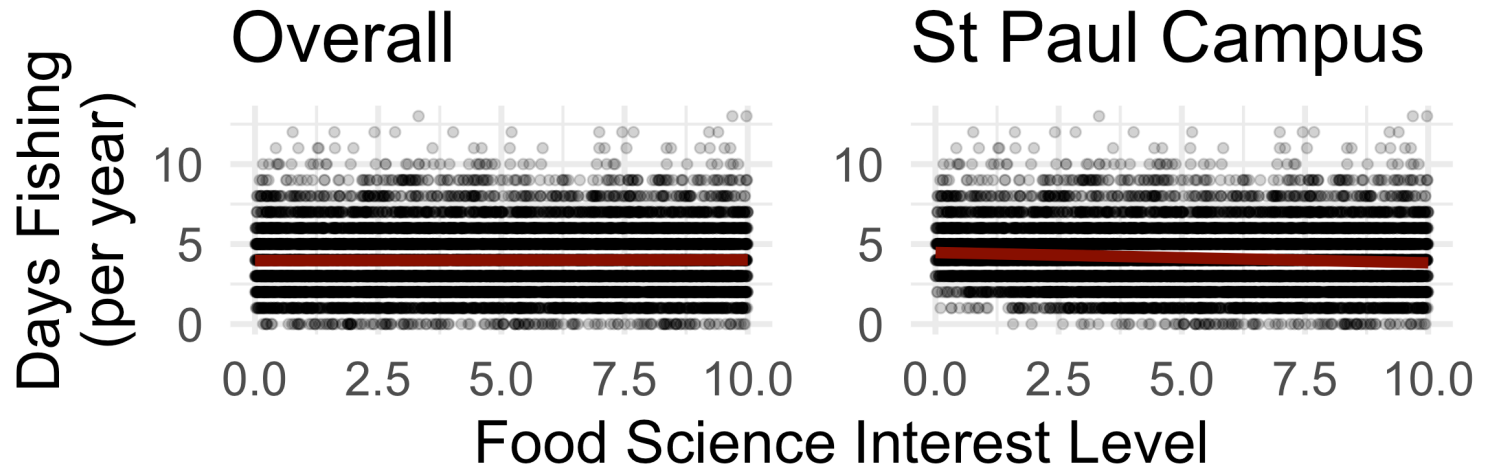
Food Science
Interest



Days Fishing

Collider

Fishing or Food -> St Paul



Example from [J. Fieberg](#)

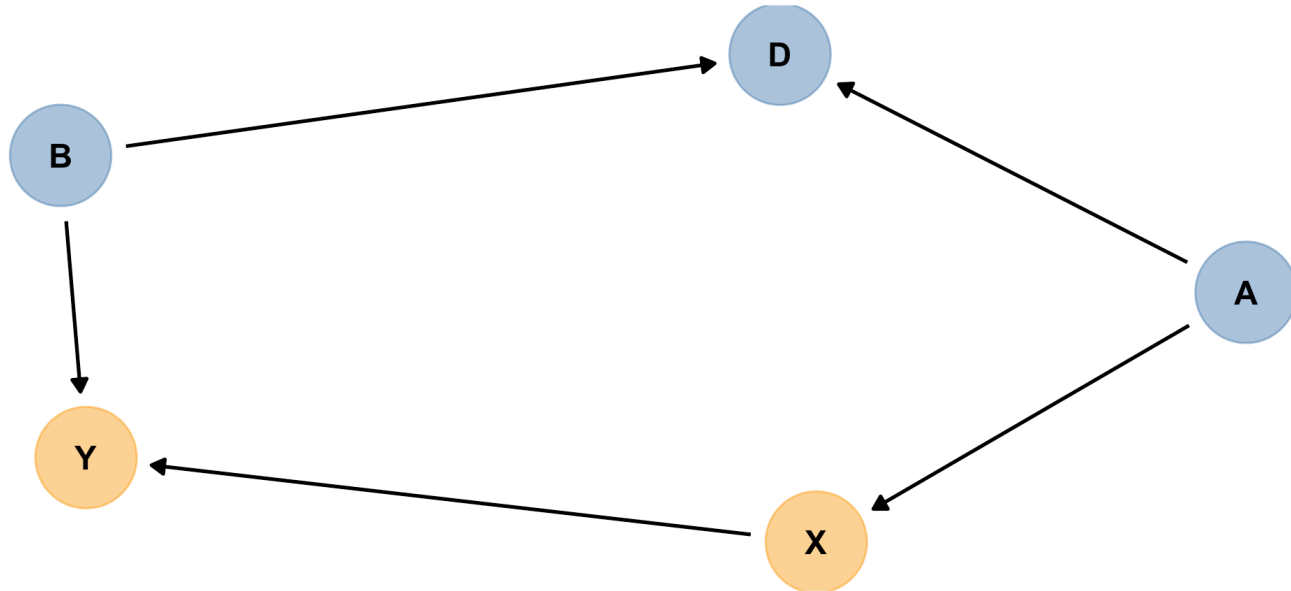
Collider

Moderator or Modifier

Also known as: *Interaction*

Can get complex

M-Bias



We've chosen some simple principles but there could be much more to learn!

Look back

**which variables should be predictors
to model HEALTH Camp test score?**

score ~

Maybe see: [Model Planning Checklist](#)

Going Further

References & Resources

- [A biologist's guide to model selection and causal inference](#) (Laubach et al.)
- *Statistics for Ecologists* (J. Fieberg) [Ch. 7](#)
- *Statistical Modeling* (D. Kaplan) [Ch. 17](#)
- [An Introduction to Causal Inference](#) (F. Dablander)
- and maybe, *The Book of Why* (J. Pearl & D. Mackenzie, ISBN: 978-0465097609)

What's Your Model Process?